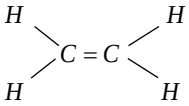
ALKENE

101. (b) Alkyne > Alkene > Alkane

102. (c) 

Ethylene sp^2-hybridization; Shape = Planar.

103. (c) Markownikoff's rule

Electrophilic addition proceeds via the formation of a more stable carbocation, which is the basis of Markovnikov's rule.

104. (b) $CH_2 = CH - Cl \xrightarrow{HCl} CH_3 - CH\begin{matrix} Cl \\ Cl \end{matrix}$

According to Markownikoff's rule *H* atom of the reagent goes to that carbon atom which is more hydrogenated.

105. (b) $CH_3 - CH = CH_2 + HBr \xrightarrow{\text{Markownikoffs rule}} \underset{\text{2-Bromopropane}}{CH_3 - \underset{Br}{\underset{|}{CH}} - CH_3}$

$CH_3 - CH = CH_2 + HBr \xrightarrow{\text{Presence of peroxide}} \underset{\text{1-Bromopropane}}{CH_3 - CH_2 - CH_2 - Br}$

106. (a) Peroxide rule is applicable only to *HBr* and not for HCl, HF and HI.

107. (c) $Cu_2Cl_2 + NH_4Cl$

108. (a) 2-chloro-1,3-butadiene

109. (a) $n\left(\underset{\text{Chloroprene}}{CH_2 = \underset{Cl}{\underset{|}{C}} - CH = CH_2}\right) \xrightarrow{\text{Polymerization}} \underset{\text{Neoprene}}{\left(-CH_2 - \underset{Cl}{\underset{|}{C}} = CH - CH_2 -\right)_n}$

110. (c) $CH_3 - \underset{CH_3}{\underset{|}{CH}} - CH_2 - Mg - Br + C_2H_5OH \longrightarrow \underset{\text{Isobutane}}{CH_3 - \underset{CH_3}{\underset{|}{CH}} - CH_3} + \underset{\text{Ethoxy magnesium bromide}}{C_2H_5OMgBr}$

111. (c) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{C}=CH-CH_3+\overset{-\,+}{HOCl}\rightarrow CH_3-CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-\overset{\overset{Cl}{|}}{CH}-CH_3$

Addition takes place according to Markownikoff's rule in which Cl^+ goes to that carbon atom which is more hydrogenated.

112. (a) In case of ethene double bond is present. Hence, addition reactions occur easily.

113. (a) $\underset{\substack{\text{Propane}\\ 1\text{ mole}\\ 42\ gms}}{CH_3-CH=CH_2}+\underset{\substack{1\text{ mole}\\ 42\ gms}}{Br_2}\rightarrow CH_3-\underset{\underset{Br}{|}}{CH_2}-\underset{\underset{Br}{|}}{CH_2}$

1, 2-dibromo propane

$\because$ 42 *gms* of propene reacts with 160 *gms* of bromine.

$\therefore$ 21*gms* of propene $\frac{160}{42}\times 21=80\ gms$.

114. (b) Butadiene $CH_2=CH-CH=CH_2$

A single bond separated by two double bonds is known as conjugated double bond.

115. (d) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH\xrightarrow[300^oC]{Cu}CH_3-\underset{\underset{CH_2}{||}}{\overset{\overset{CH_3}{|}}{C}}+H_2O$

Isobutene

116. (d) 5σ, 1π

Ethylene (C_2H_4) has:

- 4 C–H σ bonds
- 1 C–C σ bond
- 1 π bond (from double bond)

Total = 5 σ and 1 π

117. (a) Cyclic hydrocarbon in which all the carbon atoms are present in the same plane is benzene. In this $C-C$ bond length is 1.39Å which is more than 1.34A^o but less than 1.54Å. Hence bond angle is 120° with sp^2 hybridization.

118 (a) C_nH_{2n}

Alkenes contain one C=C double bond, so their general formula is C_nH_{2n}.

119. (a) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH=CH_2 \xrightarrow[ii)\ NaBH_4]{I)\ Hg(OAc)_2;H_2O} CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\underset{\underset{OH}{|}}{CH}-CH_3$

120. (b) C_3H_6 is an alkene therefore decolourizes alkaline $KMnO_4$ solution.

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